Predicting NBA Players' Three-Point Production Using Machine Learning

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GitHub link: https://github.com/110nyLaw/Team_Awesome.git

INTRODUCTION

This project investigates whether a player's past performance and game context can predict the number of three-pointers they will make in their next NBA game appearance. Using the nba_api package, we retrieved player and team game logs from NBA Stats, covering the 2015–16 through 2025–26 regular seasons. Our target variable, three-point field goals made (FG3M), is a nonnegative integer recording how many three-pointers a player makes in a single game. We frame this prediction as a supervised regression task. We plan to construct predictive features using only information available before each game, including a player's recent shooting performance, historical playing time, and opponent statistics.

The growing importance of three-point shooting in NBA offenses motivates this study. Zając et al. (2023) documented a substantial increase in three-point attempts over 40 consecutive seasons. Similarly, Wang and Zheng (2022) found that the proportion of field-goal attempts taken from three-point range rose from approximately 22% to 39% between the 2011–12 and 2020–21 seasons, while three-point shooting accuracy showed no statistically significant upward trend. These findings highlight the importance of considering both shooting accuracy and shooting opportunities when predicting three-point production. Forecasting a player's three-point production in an upcoming game using historical data could support opponent preparation and player evaluation.

Previous research has applied machine learning to basketball performance prediction. Papageorgiou et al. (2024) found that tree-based models generally performed best in their study of player performance forecasting using NBA game statistics. Nguyen et al. (2022) found that deep learning did not outperform traditional methods on their relatively small, structured dataset. Shah and Romijnders (2016) used recurrent neural networks and ball trajectory data to predict three-point shot outcomes, but their approach required observations captured during the shot. In contrast, this project uses historical box-score data to predict the number of three-pointers a player will make in an upcoming game.

DATA

We collected data through the nba_api Python package, which wraps the NBA's official statistics endpoints. All observations come from the LeagueGameLog endpoint, queried once per season at both player and team granularity. Our retrieval covers the eleven complete regular seasons from 2015–16 through 2025–26, yielding **281,163 player-game observations** across 13,209 games and 1,554 distinct players, together with 26,418 team-game records that we retain to construct opponent-context features. Each player-game record contains the player and team identifiers, game date, matchup string, minutes played, and the full counting box score, including three-point field goals made (FG3M) and attempted (FG3A). Three fields contain missing values — FG3_PCT, FG_PCT, and FT_PCT — and in every case the value is missing precisely when the corresponding attempt count is zero: FG3_PCT is absent for all 67,138

player-games with no three-point attempt, and for no others. Missingness here denotes an undefined ratio rather than an unobserved outcome, so we model the counts directly and discard the percentage fields instead of imputing them. Two further properties of the data shape our modeling choices. First, the target is a zero-inflated count: 47.4% of player-games record zero three-pointers, with a mean of 1.10 and a maximum of 14, which suggests that a count-based or tree-based model is better matched to the response distribution than ordinary least squares. Second, the endpoint reports only players who appeared in a game, so inactive players are absent rather than recorded as zeros, and 1,020 records show zero minutes after rounding. Whether a player appears at all is therefore a distinct question from how many three-pointers they make, and because the dataset includes only players who appeared in a game, our prediction task is conditional on player participation. We aim to predict how many three-pointers a player will make given that they play, rather than whether they will appear in the game.

EDA

The only files analyzed were the player_gamelogs_regular.csv, the team_gamelogs_regular.csv, player_gamelogs_regular_2015-16, and player_gamelogs_regular_2025-26. These files gave an overall understanding of the relationship between 3 pointers made and other historical data points, as well as change over the course of 11 years. Figure 1 shows the historical amount of 3 pointers made per team over the last 11 seasons. Golden State Warriors are leading this category while Orlando has the lowest 3's per game, which speaks to the individuals who have stayed and moved through the teams. Players added to this team would likely be those who can maintain these average points or would be able to boost the team average. This is important when analyzing new members to a league plus their past historical performance. Comparing the 2015-16 season to the 2025-26 season, players are attempting and making more of their shots, no matter what team they are on. There is a + 0.27 mean increase in field goals made and + 0.25 mean increase in field goal attempts, with an overall +0.03 increase in percentage over the last 11 years. This is accompanied with slightly lower rebounds (defensive and offensive) at -0.9 mean decrease, and blocks -0.02 mean decrease over the last 11 years. Although making more shots, there seems to be a slight change in how aggressive players are in stopping such attempts and requires a deeper look at how each player performs.

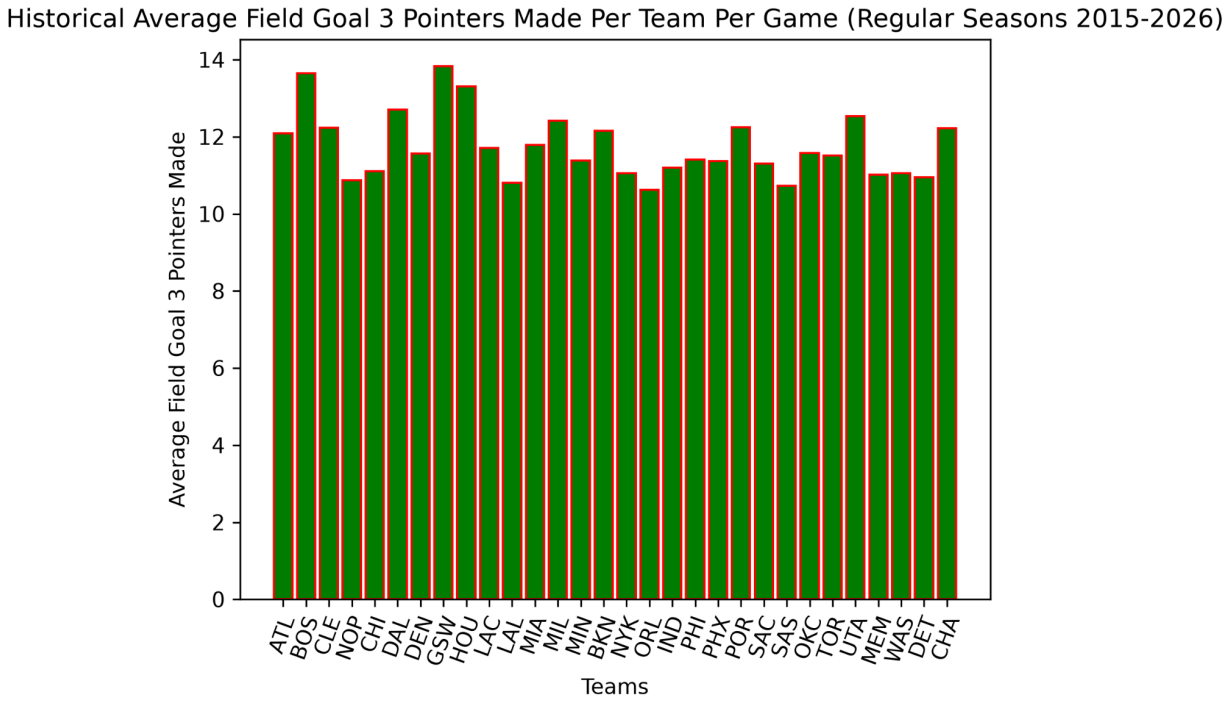
The composite player game logs shows fantasy points (.48 correlation), minutes played (.45 correlation) and assists (.26 correlation) were the top three strongest correlated factors to field goal 3's made. This is after removing columns with implicit benefits, or calculations including field goal 3's such as total points made, and columns which couldn't convert to numerical values such as game date. One such correlation was plotted with Figure 2 which shows the total number of minutes played over the last 11 seasons versus the number of field goal 3's made over the same time. It is confined to only show the most extreme cases of players who played at least one full game over the course of 11 years. There is a large fanout amongst the top 10 players and three distinct outliers from Cj McCollum, DeMar DeRozan, and James Harden who are at the ends of this fanout. Although the amount of fantasy points had a higher correlation than the minutes played, fantasy points are also an accrual of predictive worth based on previous game experience and popularity which we can not account for with the given data.

The composite team game logs shows assists (.48 correlation), minutes played (.05 correlation), and defensive rebounds (0.04 correlation) were the top three strongest correlated factors to field goal 3's

made. This also had the same columns removed to preserve .corr() input and other factors mentioned above. The correlations between these show minutes played as a common factor in procuring more 3's however are weak when compared to the team as a whole.

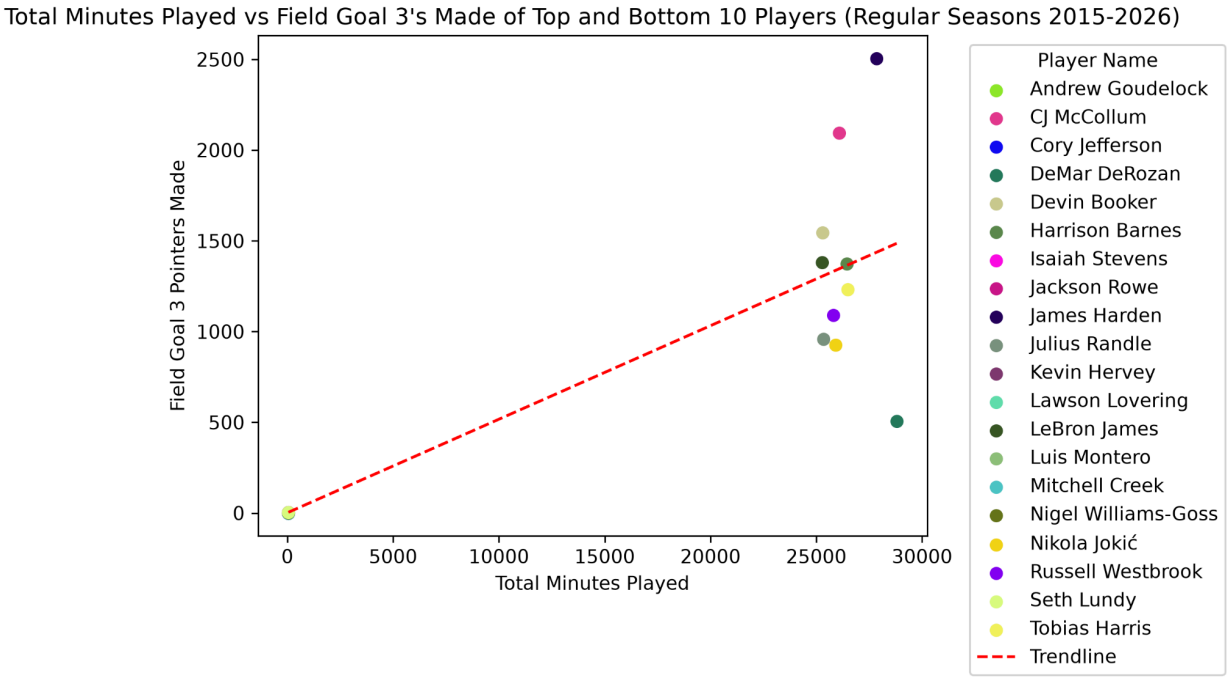
Figure 3 shows the average personal fouls per game against the average field goal 3's made per player per game. Although not one of the top correlated features, personal fouls can disrupt the actions of an active field goal being made. This also shows a slightly positive trend as the average number of fouls that occurs shows a higher fanout of average field goal 3's made per player over the last 11 seasons. The graph also has a large fanout, pointing to other features of the game effecting mean field goal 3's made. Given the increase in field goal 3's made and the decrease in rebounds and blocks historically, seeing a relationship between fouls and 3's made can be indicative in predicting future games.

Figure 1.



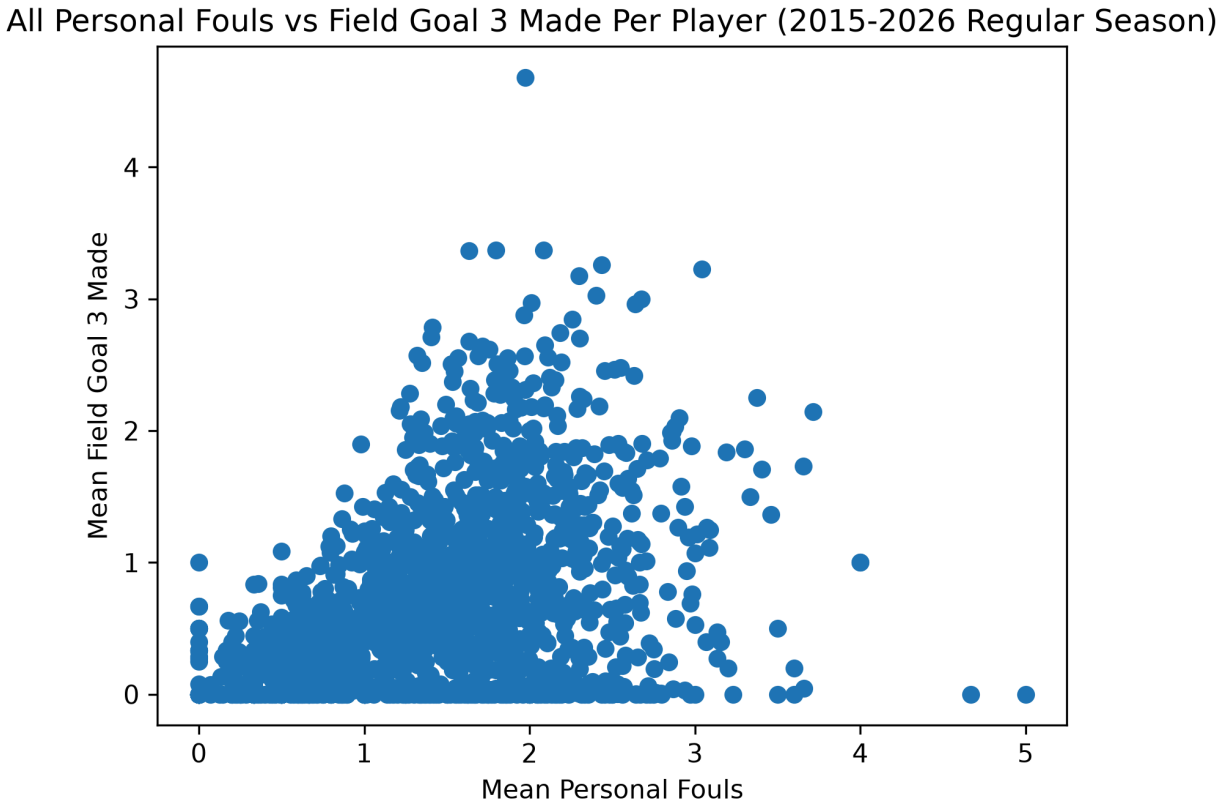
Caption: Average field goal 3 pointers made per team in a game over the 2015-2026 regular seasons.

Figure 2.



Caption: Top and bottom 10 players with total minutes played vs amount of field goal 3 pointers made. Does not account for standardization of player participating seasons. Trendline is the average minutes played to field goal 3's made relationship. Only inclusive of players who played at least one full game in minutes (48) over the 2015-2026 seasons

Figure 3.



Caption: Mean personal fouls over mean field goal 3's made per player over the 2015-2026 regular seasons. Each point is a player.

PROBLEM DESCRIPTION AND POTENTIAL IMPACT

Our objective is to predict a player's three-point field goals made (FG3M) in their next NBA game appearance. We frame this as a supervised regression problem. Predictors will be constructed from historical player and team game logs, including recent shooting performance, historical playing time, and opponent statistics. Statistics from the target game, such as actual minutes played and three-point attempts, will be excluded because they are unavailable before the game. We plan to train models on earlier games and evaluate them on later games. Since each player-game record represents an actual appearance, predictions will be conditional on the player participating.

The model could help coaches and analysts identify players expected to contribute more three-pointers in an upcoming matchup, informing defensive preparation. It could also support player evaluation by comparing actual three-point production with expectations based on historical performance and game context. However, box-score data may not fully capture injuries, lineup changes, or tactical adjustments. Predictions should therefore serve as supporting information for basketball decisions, with their practical value depending on predictive accuracy on future games.

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